



Aditya Aryan

Bachelor of Statistics (Hons.)
Indian Statistical Institute, Kolkata

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Education

Indian Statistical Institute

Bachelor of Statistics (B.Stat Hons.), Expected 2027

Jul. 2023 – Present

Kolkata, West Bengal

Kendriya Vidyalaya No. 2, Jalahalli East

12th CBSE Boards (PCM) - 95%

10th CBSE Boards - 98.8%

Apr. 2014 – Mar. 2023

Bangalore, Karnataka

Experience

Student Researcher

University of Toronto, Advisor: Prof. Xin Bing

Apr. 2026 – Present

Remote

- Working on theoretical and computational analysis of population and sample EM (Expectation-Maximization) for Gaussian mixture-type models.
- Studying update maps, fixed points, local contraction, mixture imbalance, and sampling error using tools from probability, statistics, and analysis.
- Using numerical simulations to investigate the behavior of EM iterates and support the development of results.

Data Science Intern

Ranial Systems

May 2026 – Present

New York, US (Remote)

- Developed statistical and machine-learning methods for lithium-ion battery degradation and cycle-life prediction using early-cycle voltage, temperature, capacity, and charging-protocol information, including supervised and survival-analysis approaches.
- Developed probabilistic forecasting and scenario-generation methods for NYISO electricity prices, incorporating quantile forecasts, extreme-price behavior, and uncertainty in day-ahead and real-time markets.
- Formulated battery dispatch as a constrained sequential decision problem and applied stochastic dynamic programming under state-of-charge, efficiency, cycling, and degradation constraints to study day-ahead and real-time trading strategies.

ML Research Intern

Mercity AI

Apr. – Aug. 2025

Bengaluru, Karnataka

- Developed a compact medical vision-language model by integrating medSigLIP with Qwen-2.5-7B through a Q-Former bottleneck, compressing 256 image tokens to 32 latent queries (8× compression, 70% memory reduction) while preserving probabilistic calibration in frozen-LLM caption generation.
- Trained the Q-Former with cross-entropy on MIMIC Captions; explored SFT, PPO, DPO, and GRPO using Hugging Face `trl`; extended a DeepSeek-R1-based GRPO pipeline for multimodal reasoning; increased BLEU, reduced perplexity, and published reproducible configurations and technical blog posts.

Programs & Summer Schools

exSPLORe: Explorations in Statistics, Probability, Learning and Optimization Research

2026

Saferxpress Centre for Data, Learning and Decision Sciences, Ashoka University

- Selected participant in a five-day research workshop covering probability, statistical inference, optimization, machine learning, and sequential decision-making, with tutorials and talks by researchers from Stanford, MIT, and DeepMind.

Oxford Machine Learning Summer School (OxML)

2025

- Selected for the 2025 programme, which offered MLx Fundamentals; MLx Generative AI (Agents, Alignments, Products); MLx Health & Bio; and MLx Representation Learning & Generative AI. Relevant areas included statistical learning, optimization, foundation models, Bayesian ML, reinforcement learning, computer vision, and geometric deep learning.

Summer School on Artificial Intelligence, IIIT Hyderabad

2025

- Selected for the CVIT summer school covering deep learning, CNNs, Transformers, domain adaptation, LLM reasoning, and generative modelling/diffusion; completed hands-on work with BERT fine-tuning, CLIP, and instruction tuning.

Selected Projects

Vision-Language Model Training & Reasoning (VisionLM)Independent

- Building general experimental VLM training pipelines with a custom multimodal architecture and a standard baseline to compare representation and architecture choices.
- Experimenting with GRPO-based reinforcement-learning post-training, multimodal reasoning, and reward design as an implementation-focused research testbed distinct from the medical VLM developed at Mercy AI.

Stein’s Paradox: Inadmissibility and Risk-Optimal ShrinkageSupervisor: Dr. Ayanendranath Basu, ISI

- Derived James–Stein and mean-centered shrinkage estimators under squared-error loss and extended the analysis to heteroscedastic normal means using whitening and Normal–Normal Empirical Bayes (MoM and MMLE).
- Evaluated risk using Monte Carlo simulation and Stein’s unbiased risk estimate (SURE), confirming James–Stein dominance over the MLE and the strong empirical performance of MMLE in heteroscedastic regimes.

Bivariate Copula Modelling of Extreme Air Pollution EventsSupervisor: Dr. Shyamal Krishna De, ISI

- Modeled the joint duration D and cumulative severity S of extreme AQI episodes in Bangalore (2015–2024), using empirical marginals and average-rank pseudo-observations for discrete durations.
- Fit six copulas by canonical MLE and selected Gumbel by AIC; estimated strong upper-tail dependence ($\hat{\tau} = 0.881$, $\hat{\lambda}_U = 0.881$) and $P(S > 2000 \mid D \geq 12) \approx 0.9998$.

Sequential Testing: Optimal Stopping under Reward Cost TradeoffSupervisor: Dr. Arnab Chakraborty, ISI

- Formulated a finite-horizon Markov decision problem for sequential sampling with unknown reward locations and derived a dynamic-programming recursion for the optimal stopping strategy.
- Implemented exact memoized DP and backtracking in C++; computed the optimal expected gain for $N = 20$, $C = 5$ as 7.6876 and verified agreement with the theoretical strategy.

Elo-Based Football BettingSupervisor: Dr. Ayanendranath Basu, ISI

- Built matchwise Elo features and calibrated logistic-regression probabilities for 2014–2024 EPL outcomes; evaluated forecasts using Brier score (0.2176) and log-loss (0.6249).
- Backtested a value-betting rule using model probabilities and market odds, yielding 5,139 simulated profit units from a 10,000-unit bankroll over the test season.

Relevant Coursework

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|-----------------------------------|---------------------------------|-----------------------------------|
| • Probability Theory | • Stochastic Processes | • Design & Analysis of Algorithms |
| • Statistical Methods & Inference | • Linear Algebra | • Numerical Analysis |
| • Parametric Inference | • Real Analysis | • Discrete Mathematics |
| • Linear Statistical Models | • Programming & Data Structures | • Differential Equations |

Skills

Programming: Python, C++, R, SQL.
ML & Computing: PyTorch, scikit-learn, statsmodels, NumPy, SciPy, pandas, Hugging Face Transformers, TRL, Accelerate, PEFT, bitsandbytes
Methods: Statistical machine learning, Time Series forecasting, Survival analysis, Deep learning, and LLM post-training.

Scholastic Achievements

Secured All India Rank 90 in ISI entrance examination.	2023
Secured 98.9%ile in JEE Mains (B.Tech) among 1.5 million candidates.	2023
Scored 331/390 in BITSAT.	2023
Secured All India Rank 1034 in KVPY (SA stream).	2022
Qualified for INMO (Indian National Mathematics Olympiad) merit list.	2022
Secured a Top 20 Rank in IOQM in Karnataka region.	2022
Received NTSE scholarship for being in top 1000 students among 2 million candidates.	2021
Secured Rank 72 in NTSE Stage-I.	2021